Pankayaraj . P - 4th Year PhD Student(GPA 3.77).

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Employment History

2025 (summer & fall)

Machine Learning Research Intern
Netflix USA
Manager: Ashish Rastogi
Mentor: Hafez Asgharzadeh
- Worked on pretraining LLM based embedding models

2023-2026

Research Assistant
Department of Computer Science, University of Maryland, USA
Advisor: Prof. Furong Huang.
- Worked on LLM alignment for reasoning (safety)
- Worked on robust reward modelling for LLM alignment
- Worked on RLHF poisoning in LLMs, Copyright Poisoning and Backdoor poisoning in LLMs.
- Worked on robust RAG systems

- Worked on diverse reinforcement learning.
- Works published in AAAI conference 2025, 2026 [oral], ACL 2026 and ICML, ICLR and Neurips 2024 workshops
. Some works currently under submission at , COLM 2026

2022-2022 - 7 months

Research Engineer
Singapore Management University (SMU)
Supervisor: Prof Pradeep Varakantham
- Worked on Constraint Reinforcement Learning in Hierarchical Settings
- Work was published on AAAI 2023.

2020-2021 - 12 months

Research Assistant SLTC, QBITS Lab. + **Collaborator** Flowers Laboratory, ENSTA Paris.
- Worked on the ways to improve continual offline reinforcement learning with artificial curiosity
- Work was published on Cognitive Computational Journal 2023

2019-2019 - 5 months

Research Assistant Intern
SLTC, QBITS Lab.
Advisor: D. H. S. Maithripala
- Worked on devising communication strategies for multi agent multi arm bandit problems
in both normal and delayed reward settings.
- Works were published on IEEE CDC 2020 and ECC 2020 respectively.

Research Publications

Conferences, Journals and, Workshops (Published)

1. **Pankayaraj P**, & Huang, F. (2026c). Teach a reward model to correct itself: Reward-guided adversarial failure mode discovery for robust reward modeling [**oral**], In *In the **ACL 2026**, the 64th annual meeting of the association for computational linguistics*. https://arxiv.org/abs/2507.06419

2. **Pankayaraj P**, Sehwag, U. M., Panaitescu-Liess, M.-A., & Huang, F. (2026). Advbdgen: Adversarially fortified prompt-specific fuzzy backdoor generator against llm alignment [**oral**], In ***The 40th Annual AAAI Conference on Artificial Intelligence 2026***. https://arxiv.org/abs/2410.11283

3. Panaitescu-Liess, M.-A., Che, Z., An, B., Xu, Y., **Pankayaraj P**, Chakraborty, S., Zhu, S., Goldstein, T., & Huang, F. (2025). Can watermarking large language models prevent copyrighted text generation and hide training data?, In *In the **The 39th Annual AAAI Conference on Artificial Intelligence 2025** and **3 rd NeurIPS 2024 Workshop AdvMLFrontiers** [**best paper**]*. https://arxiv.org/abs/2407.17417

4. Panaitescu-Liess, M.-A., **Pankayaraj P**, Y. K., Che, Z., An, B., Zhu, S., Agrawal, A., & Huang, F. (2025). Poisonedparrot: Subtle data poisoning attacks to elicit copyright-infringing content from large language models in the **63rd Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2025)** [**oral**], In *Naacl 2025*.

5. **Pankayaraj P**, & Huang, F. (2025). **RAGPart & RAGMask**: Retrieval-stage defenses against corpus poisoning in retrieval-augmented generation , published in the **AAAI Forntier IR workshop 2026 oral**], In ***AAAI**, 2026*. https://arxiv.org/abs/2512.24268

6. **Pankayaraj P**, Chakraborty, S., Liu, X., Liang, Y., & Huang, F. (2025). Is poisoning a real threat to LLM alignment? maybe more so than you think, In *In the **The 39th Annual AAAI Conference on Artificial Intelligence 2025** andn **ICML 2024 Workshop on Models of Human Feedback for AI Alignment***. https://arxiv.org/abs/2406.12091

7. **Pankayraj P**, Rodríguez, N. D., & Ser, J. D. (2023). Using curiosity for an even representation of tasks in continual offline reinforcement learning, In ***Cognitive Computation Journal** 2023 **Impact Factor : 5.4***.

8. **Pankayaraj P**, & Varakantham, P. (2022). Constrained reinforcement learning in hard exploration problems [**Poster**], In ***37th AAAI Conference on Artificial Intelligence** Washington, D.C. USA **Acceptance rate : 19.6 %***.

9. **Pankayaraj. P**, & Maithripala, D. H. S. (2020). A decentralized communication policy for multi agent multi armed bandit problems [[**oral**], In ***European Control Conference 2020**, Saint Petersburg, Russia **Acceptance rate : 58%***.

10. **Pankayaraj P**, Maithripala, D. H. S., & Berg, J. M. (2020). A decentralized policy with logarithmic regret for a class of multi-agent multi-armed bandit problems with option unavailability constraints and stochastic communication protocols [**oral**], In ***59th IEEE Conference on Decision and Control**, Jeju Island, Republic of Korea **Acceptance rate : 52.7%***.

11. Jayatilaka, G., Weligampola, H., Sritharan, S., **Pankayraj Pathmanathan**, Ragel, R., &], I. N. [(2019). Non-contact infant sleep apnea detection [**Presented**], In ***ICIIS 2019** Sri Lanka*.

Under Review (Current)

1. **Pankayaraj P**, & Huang, F. (2026a). *Deliberative alignment is deep, but uncertainty remains: Inference time safety improvement in reasoning via attribution of unsafe behavior to base model **Under Review**, in the **COLM 2026***.

2. **Pankayaraj P**, & Huang, F. (2026b). ***RAGPart & RAGMask**: Retrieval-stage defenses against corpus poisoning in retrieval-augmented generation **Under Review**, in the **ACL 2026***.

Symposiums (Published)

1. **Pankayaraj P**, Sumanasekera, Y., Samarasinghe, C., Elkaduwe, D., Jayasinghe, U., & Maithripala, D. H. S. (2020). *Multi-agent reinforcement learning in sparsely connected cooperative environments[**Presented**, [**best research paper**]], in **ESCaPe 2020**, Sri Lanka.*

Preprints

1. **Pankayaraj P**, Sumanasekera, Y., & Samarasinghe, C. (2019). *A review on reinforcement learning based autonomous quadcopter control.*

Education

2022-2027 — **PhD Computer Science**, Department of Computer Science, University of Maryland, USA. **GPA:** 3.77 out of 4.0
ADVISOR: **Prof Furong Huang**

2015-2020 — **B.Sc. Computer Engineering** University of Peradeniya, Sri Lanka
GPA: 3.5 out of 4.0

English Proficiency — **TOFEL**: **112**, Reading: **29**, Listening: **30**, Writing: **26**, Speaking: **27**

Academic Volunteering

2020 — **Peer Reviewer : Journal** IEEE Transactions on Communications
- Impact Factor : 5.69 (2018)

Projects

2023 — **Efficient Exploration in Reinforcement Learning** Improving the sample efficiency in RL where entropy of occupancy measure is used as an exploration mechanism
Report `https://drive.google.com/file/d/1ESQZunSYI8WegsgiQ63HJmHgUXHIF1LU/view?usp=sharing`

Virtual Maze Navigation Using Different Locomotion Techniques Analysing the effects of Redirected Walking and Steering in VR environments and proposing a hybrid locomotion technique.
Report `https://drive.google.com/file/d/11JiiJo0ZzJLbtfumzjThmq3Yp0BEG1pF/view?usp=sharing`

2019 — **Reinforcement Learning Based Autonomous Quadcopter Control**
Using Reinforcement Learning algorithms to make Quadcopter control decisions on an AirSim simulated environment

REPORT: `https://drive.google.com/drive/folders/16Ej8XL4SRrtHl58FsMMnYDaD5WYWruF9`

2018 — **A user recommendation method using Bayesian Reinforcement Learning**
Github Link : `https://github.com/pankayaraj/sitnshop/tree/BackEndAlgorithm`
REPORT : `https://drive.google.com/drive/folders/19Sq1UExeANUWQGGVf8EXBCrPRntAFvaF`

Projects (continued)

2017 **Creating a python based library with a Tensor flow back end for Bayesian Optimization and Multi Arm Bandit Problem**
GitHub Link : `https://github.com/pankayaraj/Multi-Arm-Bandit-Library`
PyPi link : `https://pypi.python.org/pypi/mabandit/1.3`
REPORT:`https://drive.google.com/drive/folders/1H2Pcbfj825LPbYjo3rnKlY0lgRQCFwXb`

2018 **SitnShop– An Advertising platform for shops** An advertising platform for anykind of shop and it also helps the customers of the shops to easily find related shops.
Github Link : `https://github.com/pankayaraj/sitnshop/`
REPORT : `https://drive.google.com/drive/folders/1ZcsJkFPDCJhvh8kOyt0LjnxBkJ1cAgaB`

Infant Sleep Apnea detection system : A portable device that can detect Sleep Apnea condition in infants using techniques such as Optical flow, Edge detection, Fourier analysis with python, Raspberry pi.
Github Link : `https://github.com/pankayaraj/Sleep_Apnea_Detection`
REPORT:`https://drive.google.com/drive/folders/17fLXhj1uxl5MuqNqEM46_tYuRsWoXC2Z`

Making a central server for the sleep apnea problem
Technologies : Python, Django, Django rest framework, HTML/CSS,JS
Github Link: `https://github.com/pankayaraj/Django_Server_Sleep_Apnea`

2017 **ExpertMiner :**
Earth resource location prediction using Hyper Spectral Images from satellites Techniques : Pattern Recognition, Correlation Mapping
Github Link : `https://github.com/pankayaraj/HSI_Project`

Skills

Languages	English (**TOFEL**: **112**, Reading: **29**, Listening: **30**, Writing: **26**, Speaking: **27**), Tamil(Native).
Programming	Python , LaTeX, C++,
Libraries	Tensorflow, Pytorch, Kivy, Numpy, Scipy
Web Dev	Dijango

Miscellaneous Experience

Awards and Achievements

2017 **ACES Hackathon 2017(Intra university hackathon)** : Project:Expert miner: Softwaresection winners, Best idea of the competition

2016 **ACES Coders v6.0** (Inter university programming competition):Country Rank : 4th

IEEExtreme 10.0 Programming competition(24 hour Global Programming competition
Country Rank: 38th
World Rank: 330th

References [In the order of employment (recent to furthest)]

Prof Furong Huang
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